

MOHAMMAD KAMIL

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Computer Science Undergraduate | Vellore Institute Of Technology, Bhopal | Address: Jhansi, 284002

SUMMARY

- Computer Science and Engineering undergraduate at VIT Bhopal (CGPA: 9.02) with strong foundations in data structures, algorithms, and computer science fundamentals. Proficient in Python and Java, with hands-on experience in SQL, relational databases, and AWS cloud services. Experienced in building and deploying AI/ML systems — including a medical CT scan classifier (91.3% accuracy) and a multi-module precision farming platform (94.38% disease detection accuracy). Eager to leverage core CS knowledge, programming skills, and database expertise to solve impactful, real-world problems.

EDUCATION

Vellore Institute of Technology (VIT) September 2023 – August 2027
B.Tech – Computer Science and Engineering CGPA: 9.02 / 10.0
Relevant Coursework: **Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Artificial Intelligence, Machine Learning, System Design**
Madhya Pradesh, India

TECHNICAL SKILLS

- CS Fundamentals (Core Strength):** Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming (OOP), System Design, Complexity Analysis
- Programming Languages (Core Strength):** Python (primary), Java — applied in production ML systems, data pipelines, and academic projects; C++, Assembly Language
- Databases & SQL (Core Strength):** SQL (complex queries, joins, indexing, optimization), Oracle DBMS, schema design, ER diagrams, normalization (1NF–3NF), relational data modeling
- Cloud – AWS & GCP:** AWS (EC2, S3, IAM — exposure); Google Cloud Platform (GCP) — VM instances, load balancers, scalable architecture deployment across 3 regions
- Machine Learning / Deep Learning:** Keras, Convolutional Neural Networks (CNN), Transfer Learning, Hyperparameter Tuning, Feature Engineering, PCA, Recursive Feature Elimination (RFE)
- Computer Vision:** OpenCV, Image Classification, Data Augmentation, Batch Normalization, Dropout Regularization
- Data Science Libraries:** Pandas, NumPy, Matplotlib — data wrangling, visualization, and statistical analysis
- Developer Tools:** Git, GitHub, Visual Studio Code, Tkinter (GUI Development)

PROJECTS

•Kidney Disease Prediction System – Medical AI Image Classifier

August 2025

- Developed a deep learning classifier in **Python** for 12,000+ labeled CT scan images across 4 categories (Cyst, Tumor, Stone, Normal) using a custom Convolutional Neural Network (CNN) built with Keras.
- Engineered a GPU-optimized data pipeline with 256x256 image resizing and batch size 64, maintaining 80%+ GPU throughput throughout training — applied core **algorithms** and **data structure** principles for efficient batch processing.
- Architected CNN with Conv2D, MaxPooling, BatchNorm, Dropout, ReLU, and Softmax layers (2.3M+ parameters); achieved 91.3% training accuracy and 89.6% validation accuracy across 20 epochs.
- Reduced total training time by 24% by integrating early stopping and real-time image augmentation, effectively preventing overfitting on limited medical data.
- GitHub:** github.com/kamil-ctr/kidney-disease-classifier

•Precision Farming System – End-to-End AI Agricultural Advisory Platform

July 2024

- Architected a full-stack AI platform in **Python** combining three independent ML modules — Fertilizer Recommendation, Crop Prediction, and Plant Disease Detection — into a unified advisory system for Indian farmers.
- Curated and processed 84,000+ agricultural records across 9 soil and environmental features from 18 Indian states; leveraged **SQL**-based data querying and feature validation pipelines; fertilizer model achieved +/-5 NPK deviation, cutting over-application by 37.8%.
- Boosted crop prediction accuracy to 91.4% by applying PCA and Recursive Feature Elimination (RFE), reducing feature dimensionality by 42% across 30+ validated crop classes — demonstrating strong **algorithmic thinking** and optimization.
- Trained a CNN on 32,000+ plant images spanning 38 disease classes; attained 94.38% test accuracy via systematic hyperparameter grid search across 96 configurations.
- Deployed all three models in a Tkinter-based GUI with sub-1.5-second response times; field-validated on 150+ cases with 97% user satisfaction.
- GitHub:** github.com/Nishesh1708/Farmer-Dashboard

•Student Record Management System – Java/C++ Console Application

January 2024

- Designed and implemented a file-based CRUD system using **OOP principles** (encapsulation, abstraction, inheritance) managing 1,000+ student records, with structured class validation to enforce data integrity — directly applicable to **database-driven** application design.
- Applied core **data structures** (linked lists, arrays, file I/O) and **algorithmic** logic for search and retrieval; cut manual entry errors by 85%+ through strict input validation.
- Delivered 15+ modular milestones, improving overall maintainability and team efficiency by 20%.
- GitHub:** github.com/kamil-ctr/student-record-system

•Matrix Operations Toolkit – Generic C++ Template Library

September 2023

- Authored a reusable C++ library for 3 core matrix operations using templates and operator overloading, enabling type-agnostic computation — demonstrating strong grasp of **CS fundamentals** and **algorithmic optimization**.
- Optimized algorithmic logic to improve matrix computation speed by 30%; correctness verified across 50+ diverse test matrices.
- GitHub:** github.com/kamil-ctr/matrix-toolkit

•Array Utilities Module – Data Structures Library

July 2023

- Built 6+ modular array utilities — including sort, reverse, binary search, and frequency count — applying core **data structures & algorithms** knowledge; verified across 25 constraint-varied datasets and arrays of up to 10,000+ elements.
- Achieved 100% functional accuracy with consistent time complexity; emphasized clean, readable code and reusable component design.
- GitHub:** github.com/kamil-ctr/array-utilities

CERTIFICATIONS

•Oracle – Database Management Systems (DBMS)

2024

- Scored 80%; completed 20+ hands-on **SQL** exercises, 10 Entity-Relationship (ER) diagrams, and normalization tasks across 15+ complex relational schemas. [Certificate]

•Oracle – Generative AI and Prompt Engineering

2024

- Scored 88%; designed 5 high-precision AI prompts achieving 35% output improvement; completed 12+ labs on LLM fine-tuning, few-shot prompting, and chain-of-thought reasoning. [Certificate]

•Google Cloud – Fundamentals of Cloud Computing (GCP)

2023

- Gained hands-on experience with 8+ GCP services; deployed VM instances and load balancers across 3 regions; simulated cloud architectures that reduced provisioning time by 40%. Concepts transferable to **AWS** cloud infrastructure. [Certificate]

POSITIONS OF RESPONSIBILITY

- Hostel Discipline Committee Member** – Upheld student welfare policies and mediated conflict resolution across hostel facilities.
- Event Volunteer** – Coordinated logistics, scheduling, and team communication across technical, sports, and cultural college festivals.
- Hackathon Participant** – Competed in Android Club and Microsoft-organized hackathons, collaborating under time constraints to build functional prototypes.